Agatha Ulina Silalahi

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Summary

Data Scientist with a Master of Mathematics (Data Science) from Universitas Indonesia and NLP research experience at Bank Indonesia Institute. Works end to end on data for financial services: SQL analysis and dashboards, credit risk and fraud models, and LLM applications. Evaluates models before release with baselines, temporal validation, calibration and release gates, and LLM apps with hand-checked golden sets, guardrails and cost limits. A former mathematics teacher who explains results clearly to non-technical audiences; every project metric below can be re-run from a public repository.

Education

Universitas Indonesia
Master of Mathematics, Data Science specialization (GPA: 3.61) August 2023 – Oct 2025
Relevant Coursework: Computational Data Science, Advanced Machine Learning, Applied Deep Learning, Advanced Statistical Analysis, Advanced Data Exploration and Visualization, Advanced Bioinformatics, Advanced Genomic Data Science.

Pacmann Academy - Non Degree Program 2023 – 2025
Job Preparation Program Basic: AI/ML Engineer
Relevant Coursework: Python Programming & Software Engineering, Probability & Statistical Testing, Linear Algebra, Machine Learning Algorithms, APIs & Deployment, Recommender Systems, Neural Networks, Machine Learning Product Management, Natural Language Processing.

DQLab - Bootcamp
Data Analyst with SQL & Python in Google Platform, Batch 8: SQL, Python, Looker Studio storytelling, and an ARIMA forecasting capstone on the Superstore dataset.

Universitas Negeri Medan
Bachelor of Mathematics Education, FMIPA (GPA: 3.53) 2017 – 2021

Work Experiences

Bank Indonesia Institute (BINS) | Data Scientist Research Assistant Mar 2025 – Present
- Build NLP pipelines that score Bank Indonesia press releases and related news for clarity, sentiment and coverage, for research on central bank communication.
- Combine lexicon-based scoring, machine learning classifiers and manual annotation to label sentiment in financial news.
- Use topic modelling and time series analysis to track inflation and exchange-rate themes, as input to an early warning indicator.

PT Bank Rakyat Indonesia (Persero) Tbk | Artificial Intelligence Research Intern Aug 2024 – Jan 2025
- Built a prototype LLM resume parser: an internal LLM prompted with a fixed JSON schema at temperature 0 extracts name, contact, education, skills and work history from CV text, served through a Flask web app.
- Turned unstructured CV text into structured records that HR could filter and compare during screening.
- Public version of the prototype with synthetic data: GitHub.

SMA Global Mandiri Cibubur | Mathematics Teacher Jun 2022 – Jun 2023
- Taught mathematics, prepared annual and semester programs, syllabi and lesson plans, and coached students in academic and non-academic activities.

SMPK Penabur Gading Serpong | Bilingual Mathematics Substitute Teacher Jan 2022 – Apr 2022
- Managed the class and taught mathematics bilingually (English and Indonesian).

Private Teacher | Mathematics Tutor 2020 – Present

Project

- **NLP/LLM Engineering** | Project · Live Demo | **FinSight IDX: RAG Question Answering over Annual Reports** — Python, Claude API, BM25, ChromaDB, Tesseract OCR, Gradio, Docker
  - RAG app that answers questions about the 747-page BCA 2023 annual report and cites the page of every figure.
  - Built a golden set of 58 questions with answers and pages checked by hand in the PDF, then measured retrieval (recall@k, MRR) and answers (accuracy, citation, refusal) on it.
  - Audited my own index and found duplicate chunks, wrong page numbers and text skipped by the chunker; fixed them with regression tests and a versioned index manifest.
  - Indexing each financial-table row as its own chunk raised hybrid page recall@5 from 0.30 to 0.58. BM25 alone did better on this document, so the app uses BM25 with 10 chunks (recall 0.70).
  - Answers are correct for 79% of questions (65% before the retrieval change), cite the right page 91% of the time, and refuse every question the report cannot answer, at about one US cent per question.
  - Serving: input validation, per-visitor and global rate limits, a daily cost ceiling, telemetry without storing question text, and a smoke test for the live Space; 220 tests in CI.
- **Machine Learning** | Project | **Credit Default Early Warning System with Explainability** — Python, XGBoost, SHAP, scikit-learn

  - Default model on 307,511 Home Credit applications (8.1% default rate): XGBoost ROC-AUC 0.763 and PR-AUC 0.251, against 0.745 and 0.225 for an imputed, scaled logistic regression baseline.
  - The top 20% of scores captures 52% of defaults; probabilities are calibrated so they can feed expected-loss estimates.
  - Removed gender from the features and measured the cost in accuracy; SHAP shows two external bureau scores carry 29% of the model's attribution. Built a PSI drift detector and tested it on simulated drift.
- **Credit Risk Modelling** | Project · Dashboard | **CreditLens: Credit Scoring Research with a Release Gate** — PostgreSQL, dbt, XGBoost, FastAPI, Streamlit
  - Warehouse of 2,260,668 LendingClub loans in PostgreSQL and dbt, with a temporal train, validation and test split fixed before modelling.
  - Compared a constant baseline, logistic regression and XGBoost on the same cohort. The best model reached average precision 0.220 on the frozen test, below the 0.25 gate set in advance, so I held it back from release.
  - The FastAPI scoring service refuses unreleased models; a public Streamlit dashboard shows only aggregates; 158 automated tests.
- **Data Engineering** | Project | **End-to-End Data Pipeline for Credit Risk ML** — Python, PostgreSQL, XGBoost, Docker, Kubernetes, GitHub Actions
  - Five-stage ETL from raw CSV to model features for 150,000 borrowers (Give Me Some Credit), producing identical metrics on PostgreSQL and SQLite.
  - XGBoost ROC-AUC 0.866, PR-AUC 0.394 and KS 0.579, against 0.857 and 0.360 for logistic regression; early stopping uses a validation split, not the test set.
  - Found that the World Bank macro join added no information (the dataset has no dates), proved it with unique-value counts and an ablation, and removed those features. Docker, a Kubernetes Job on local minikube, 11 tests and CI.
- **Data Analytics** | Project · Tableau · Looker Studio | **NusaCommerce E-Commerce Analytics Dashboard** — PostgreSQL, SQL, Python, Tableau Public, Looker Studio
  - SQL-first analysis of 20,848 orders from one Shopee seller (Dec 2023 – Nov 2025): COD is 55% of orders but 38% of revenue; 13.6% of orders are cancelled.
  - RFM segmentation of 409 buyer locations (the data has no buyer ID): 88 Champion locations generate 71% of revenue. Executive dashboard in Tableau and operational dashboard in Looker Studio, refreshed in October 2026.
- **Fraud Detection & MLOps (in progress)** | Portfolio · BAF Paper | **SIGAP: Fraud Scoring with an Analyst Copilot** — PySpark, Airflow, MLflow, XGBoost, FastAPI, MCP, LangGraph, Docker
  - Fraud scoring for online bank-account applications on the BAF dataset (Feedzai, NeurIPS 2022: 1 million applications, 8 months), following the paper's protocol (month-based splits, TPR at 5% FPR, FPR parity by age group), plus a copilot that explains each score to analysts.
  - Now: a skeleton running end to end on synthetic BAF-shaped data (PySpark medallion pipeline with a star schema, training, monthly replay, FastAPI, MCP copilot); 39 tests.
  - Next, by November 2026: leakage-safe features, XGBoost vs a logistic baseline with a release gate and MLflow, fairness on the six BAF variants, PSI monitoring with Airflow, copilot guardrails and golden-set evaluation, and a public demo; results reported only once measured.
- **Sentiment Analysis** | Project | Fine-Tuning a Pretrained BERT Model for Text Classification
- **Data Science Computation** | Project | Prediction of Medical Insurance Cost with an XGBoost Model
- **Data Science Statistics** | Project | Comparison of stress causes in developed and developing countries using clustering and multinomial logistic regression on the COVID Stress Global Survey dataset
- **Data Analyst Bootcamp DQLab** | Project | Time series forecasting of Superstore sales with an ARIMA model

Skills

- **LLMs & Retrieval:** Claude API, prompt and output-schema design, retrieval-augmented generation (BM25, embeddings, ChromaDB, reciprocal rank fusion), PDF extraction and OCR, golden-set evaluation (recall@k, MRR, answer and citation accuracy).
- **Credit Risk & Evaluation:** ROC-AUC, PR-AUC, KS, calibration, temporal validation, bootstrap confidence intervals, PSI drift monitoring, release gates, removing protected attributes.
- **NLP:** Sentiment analysis, topic modelling, BERT fine-tuning, Indonesian financial text.
- **Data Analysis & BI:** SQL (CTEs, window functions), cohort (vintage) and RFM analysis, Tableau Public, Looker Studio, matplotlib, seaborn, ggplot2; explaining results to non-technical audiences (former mathematics teacher).
- **Programming:** Python (pandas, NumPy, scikit-learn, XGBoost, LightGBM, SHAP, SQLAlchemy) and R.
- **Data Engineering:** SQL on PostgreSQL, dbt, ETL pipelines, REST API ingestion, export to Google BigQuery; PySpark (medallion architecture, partitioning) in progress through SIGAP.
- **MLOps & Engineering:** Git and GitHub Actions, pytest, Docker, FastAPI, Gradio, Streamlit, Hugging Face Spaces, Kubernetes on local minikube, rate limiting and cost control; MLflow and Airflow in progress through SIGAP.
- **Statistical Analysis:** Hypothesis testing, regression analysis, clustering.

Certificate

Fundamental of Probability & Random Variables and Statistical Test | Pacmann Certificate
Bootcamp Data Analyst with SQL & Python in Google Platform Batch 8 | DQLab Certificate